Characteristics of the CPU

Introduction

There are a number of common characteristics of a CPU. For your exam, you need to understand what these characteristics are and how they affect the performance of a computer. The next few pages will introduce these characteristics and explain their importance.

Clock Speed

The speed of the Fetch-Decode-Execute cycle is determined by the CPU's clock chip. This chip uses a vibrating crystal that maintains a constant rate. The speed of the clock is measured in hertz (Hz) which is the amount of cycles per second. A clock speed of 500Hz means 500 cycles per second. Current computers have CPU clock speeds of 3GHz which means 3 Billion cycles per second.

Overclocking

It is possible to increase the clock speed for a CPU. This is known as overclocking. In theory, if the clock is faster than the CPU can perform more calculations and perform faster.

The problem is that CPUs get hotter the more work they do – so overclocking is dangerous without the appropriate heat management.

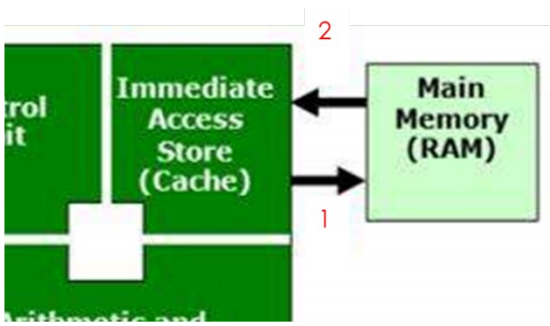
The Cache

As we have just seen CPUs can work very quickly indeed but unfortunately CPUs can only work when supplied with data. The RAM (that supplies the data) cannot work at the same speed. To overcome this the CPU's cache memory will not just copy the instruction needed at that time, instead it will also copy the continuing instructions. Cache memory has read speeds similar to the CPU and is therefore much faster than RAM.

So, to improve efficiency the CPU's 'Control Unit' will look first in the cache for the next instruction to see if it has already been copied which reduces the time taken to access data. If the cache is larger, it is more likely that the next required instruction has already been transferred from the RAM to the CPU thus improving process time.

1. If the required data is not in the cache, the control unit will request it from RAM

2. Data/Instructions (and future instructions) copied to cache for quicker access



CPU Cores

Computer Science UK Membership Site Licence: Do not share outside your centre.

CPUs of today are pretty much at the limit of today's technology. So if CPUs can't go faster, the solution its to add 2 CPU chips, or 4, or 8, etc). These chips are called 'cores'. Because the various cores can each carry out their own Fetch – Decode – Execute cycle it means that instructions can be processed at the same time. Allowing a CPU to process MORE data during the same time period.

There are two ways in which a CPU with more than one core can process more instructions in the same time period:

- They can carry out Parallel Processing which is when the same program can have two instructions processed at the same time.
- More cores also enables Multi-Tasking where each core can each process two different programs' instructions at the same time.

Either way, more cores means more instructions being processed at once. However, it doesn't necessarily mean CPUs work faster as some programs do not allow more than one instruction to be processed at once, so be careful when answering exam questions on this!

Embedded Systems

For your exam, you will also need to understand what an embedded system is. When we think of a computer, you usually think of a PC and as many of you know, a PC is made up of various components including a motherboard, CPU, RAM, input devices etc. But of course a computer is any programmable machine...or any electronic device which takes in data, processes it and then outputs the result.

So when you consider devices like cameras and watches, as these are programmable machines, they can also be called computers. The main difference is that these computers run specific tasks – they are not general purpose. Because of this, they do not need to have separate components as these devices won't need updating when new software / hardware is released. These systems instead have all of their components arranged together on a single circuit board.



Source:
http://neuronelab.unisa.it

As a result they are known as embedded systems as all of their hardware is embedded together as one.

Questions *(The question zone you choose must either match your target grade or be higher!)*

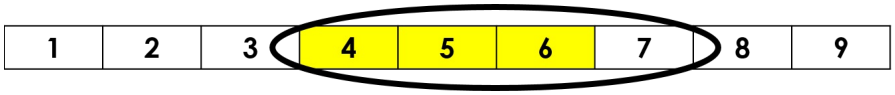
1	2	3	4	5	6	7	8	9

Computer Science UK Membership Site Licence. Do not share outside your centre.

Question Zone 1-3

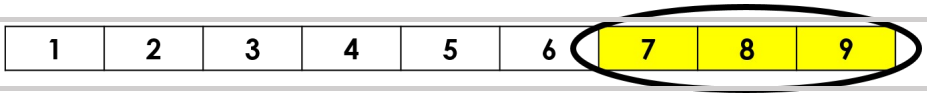
1. State what is meant by the CPU's clock speed and what is meant by the term overclocking. [3]
2. Describe what is meant by the term 'cores', when it comes to CPU architecture. [2]
3. Describe the role of cache in a computer system. [3]

Question Zone 4-6



1. Describe the role of cache in a computer system. [3]
2. Explain the effects that a higher clock speed will have on the performance of a CPU. [4]
3. Explain the term 'parallel processing' in relation to CPU cores. [4]

Question Zone 7-9



1. Explain, using examples, the term 'parallel processing' in relation to CPU cores. [4]
2. Explain the need for cache memory and the reason why larger cache will improve performance. [5]
3. Discuss the accuracy of the statement "A CPU with more cores will always outperform those with fewer". [5]

Computer Science UK Membership Site Licence: Do not share outside your centre.

Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms.

Cache: *A relatively small amount of memory located next to the CPU, used to supply the CPU with regularly required instructions / data at speed.*

Clock Speed: The amount of F-D-E cycles that can be performed per second.

CPU Cores*: An independent processing unit that is able to fetch, decode and execute instructions.*

Computer Science UK Membership Site Licence: Do not share outside your centre.

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a key word.
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**!

Computer Science UK Membership Site Licence: Do not share outside your centre.

Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

_____

_____

_____

_____

Computer Science UK Membership Site Licence: Do not share outside your centre.